International Journal of Obstetric Anesthesia 65 (2026) 104828

Contents lists available at ScienceDirect

International Journal of Obstetric Anesthesia

journal homepage: www.elsevier.com/locate/ijoa

Editorial

Intraoperative pain during cesarean delivery: reflections and next steps

ARTICLE INFO

Keywords:
Intraoperative pain
Cesarean delivery
Research
Obstetric anesthesia

Despite advances and optimization in neuraxial techniques for cesarean delivery, recent reports show that approximately 15% of women experience pain during cesarean delivery under neuraxial anesthesia.[1] With more than 1.16 million cesarean deliveries performed each year in the United States,[2] this problem affects approximately 170,000 women annually. National societies including the American Society of Anesthesiologists (ASA) and the Society for Obstetric Anesthesia and Perinatology (SOAP) as well as lay press including the New York Times,[3–5] have highlighted pain during cesarean delivery as a priority area for research and for improvement of clinical management. The 2024 ASA *Statement on Pain During Cesarean Delivery* recognizes intraoperative pain management as a quality metric.[3] In Europe, the Obstetric Anaesthetists' Association (OAA) and the French Practice Bulletin published recommendations for prevention, recognition, and follow-up of patients who experience intraoperative pain management during cesarean delivery.[6,7]

Pain during cesarean delivery research has predominantly been evaluated with quantitative methodology with most focus on reporting the incidence of the problem.[1,8] Emerging qualitative research in this field was identified as a knowledge gap needed to bridge incidence and patient experience.[9,10] In the recent qualitative study by Callihan et al. published in the *International Journal of Obstetric Anesthesia*,[11] we are reminded that pain experienced by women during cesarean delivery is not only a physiological phenomenon but also a profoundly emotional one, that can potentially shape postpartum recovery, mental health and future decisions regarding childbirth.[11] Descriptions and intense narratives of fear, loss of control, pain endurance, and trust in healthcare providers that accompany patients' intraoperative pain experiences, suggest that these experiences are neither benign nor isolated in nature. Together, these data signal the need for the next phase in obstetric anesthesia research, one placing measurement and optimization of patient experiences at the center alongside research and clinical initiatives that improve *preventing, identifying and treating* intraoperative pain during cesarean delivery.

The quantitative prevalence lens

Early work, such as single center prospective and retrospective studies from the United Kingdom and United States, identified the prevalence and risk factors for inadequate anesthesia, which included delivery urgency, epidural top-up and provider inexperience.[12,13] However, many of the observational cohorts which explored intraoperative pain were limited by their single center, retrospective study design, inconsistent definitions for inadequate neuraxial anesthesia and intraoperative pain, and notably the lack of patient-reported outcomes. In fact, most studies used clinician interventions as a surrogate marker for pain, (such as intravenous supplementation, neuraxial medication or replacement, or general anesthesia), rather than patient-reported pain.

While contemporary randomized controlled trials (RCTs) are broadly associated with improved methodological rigor, they may not be generalizable or applicable to the evaluation of routine anesthesia care. A systematic review of 54 RCTs published in 2022 reported that 14.6 % of women received supplemental analgesia or conversion from neuraxial to general anesthesia during scheduled cesarean delivery performed under neuraxial anesthesia. In this study conversion to general anesthesia was rare, occurring in only 0.06 % of cases.[8] These findings were supported by high levels of evidence and demonstrated a low risk of bias due to strict inclusion criteria by assessing only RCTs. However, this incidence may have been inflated by the "Hawthorne effect," since patients participating in prospective trials can receive heightened attention including anesthesia practices of block testing, and frequency of questioning regarding intraoperative pain symptoms, and general care, which may not be typical of routine patient management. Regardless, the study reported a pooled incidence using the highest-quality data available and demonstrated that intraoperative pain is a common complication that warrants appropriate counseling and heightened attention from anesthesiologists.

What remained missing was the patient-centric prevalence: the number of women who experience pain that they themselves identify as significant, rather than pain inferred from clinician interventions, such

https://doi.org/10.1016/j.ijoa.2025.104828
Accepted 3 December 2025
Available online 5 December 2025

as supplemental analgesia or conversion to general anesthesia, provided by the anesthesia team or records. A systematic review of randomized and non-randomized studies tried to address this knowledge gap. The pooled incidence of patient-reported pain during cesarean delivery performed under neuraxial anesthesia from 34 studies (n = 2,061) published between 1990 and 2023, from 20 different countries, was 17 %.[1] Interestingly, this incidence was remarkably similar to the previous systematic review that only assessed interventions.[8] However, to determine prevalence, quantitative definitions (whether interventions or patient-reported pain) usually use a dichotomous ("yes/no") outcome as it relates to pain during cesarean delivery. While convenient, such binary outcomes fail to capture the range of experiences, and the emotional and psychological dimensions of intraoperative suffering. This raises the topic of how parturients can be best prepared emotionally, prior to a scheduled or emergency cesarean delivery. This could be addressed with the development of assessment tools for these aspects of perinatal care by anesthesiology, obstetric and nurse teams.

In addition, studies have shown that anesthesiologists and obstetricians are unreliable at identifying patients who experience pain during a cesarean delivery, when asked immediately after surgery.[14] Furthermore, there may be a mismatch between medication administration and actual intraoperative pain,[15,16] therefore evaluating interventions in response to presumed pain may be misrepresenting the true incidence, significance and impact of pain. These recent findings underscore the need for a phenomenological assessment as described in other settings.[17,18] For our purposes, this would involve patients being asked about their comfort, sensations and experience of pain during cesarean delivery. In other words, relying on the 'objective' data derived from the anesthesia records, rather than the 'subjective' experience provided by patients, may incorrectly estimate the prevalence of pain during cesarean delivery.

The qualitative experience lens

A qualitative study published in 2024 provided a descriptive summary of surgical sensations experienced during cesarean delivery in patients that did not experience intraoperative pain.[19] In the qualitative concept-elicitation study by Callihan et al.[11] performed at a single academic institution in the United States, the authors begin to add to the body of qualitative work of patients' experiences of pain during cesarean delivery. Ten women who experienced intraoperative pain during cesarean delivery under neuraxial anesthesia participated in an interview within 72 h of delivery, using a semi-structured interview guide. Interviews were digitally recorded, transcribed and then thematic analysis was performed using an iterative inductive approach. Nine thematic domains emerged during the interviews: maternal psychological state, patient expectations, anesthesia care in labor and delivery, staff interactions, patient impression of the anesthesia provider, anesthesia care in the operating room, obstetric variables, testing of the block, and sensations during surgery.

Patients described not only physical pain but also existential fears: *"I wanted to be alive when she was born,"* and *"I needed to hear her cry before I was knocked out."* The desire to endure pain to ensure the baby's safety, the need to trust their anesthesia team, and the profound influence of the operating room environment on patient experiences, were recurrent themes described by the women. These descriptions suggest that intraoperative pain experiences are more than a physiological sensory transmission but rather represent a complex interaction of psychological constructs including anxiety, expectations of surgery, and perceived safety for both maternal and infant life, in the most unique setting for the patient undergoing surgery awake and their support person. Although domains such as "interactions with staff" and "impression of anesthesia provider" seem intuitive, they have previously not been explicitly identified to be important in this context and seem inherently difficult to consistently evaluate in a time efficient manner, especially in intrapartum or emergent settings. Measures to assess these constructs or domains are currently lacking but may prove to be crucial to improving our understanding of patient satisfaction with anesthesia care, recall of events, and psychological recovery following pain experiences.

When and how should we ask about pain?

The European Society of Anaesthesiology and Intensive Care (ESAIC) provides guidance for managing a failing epidural during labor (assess, identify, intervene, escalate, communicate and document), which may help to reduce intraoperative pain through prevention.[20] However, a remaining challenge for researchers and clinicians who conduct prospective studies and quality improvement initiatives is deciding *when* and *how* to enquire about intraoperative pain. Communication skills are rarely taught to anesthesia team members, either at a team or at a hospital scale, and such training may be effective at improving quality of care and intraoperative pain management.

Repeated intraoperative questioning at set intervals e.g. "Do you have pain?" or "Are you in pain?" asked every five minutes, may inadvertently negatively impact patient confidence in their provider or even induce pain through suggestion. Continuous verbal intrusions by clinicians may also negatively impact maternal-neonatal and family bonding. Furthermore, the optimal frequency for repeated questioning is unknown, and may differ for individual patients depending on the circumstances surrounding delivery. This strategy may also be limited by loss of valuable patient feedback regarding their experiences in between the timed questions. Conversely, not explicitly enquiring about pain may risk non-identification of a distressed patient who may feel unable or frightened to speak up or verbalize their pain experience. Patient preferences would ideally be established prior to surgery; however emergent surgery may result in the inability for providers to establish patient preferences.

Framing of pain questions by anesthesia providers is also important. For example, stating "If you feel any discomfort, please tell us," may be more likely to empower patients than alarm them. Negative words such as pain may induce nocebo effects and worsen the pain experience or reported pain.[21] Differences in how pain during cesarean delivery is elicited (e.g. pain score versus comfort score) may also impact pain experiences and satisfaction with anesthesia management.[22] The optimal technique for patient enquiry by anesthesia providers requires further study.

Alternative approaches include deferred questioning until after surgery, in the post-anesthesia care unit or following admission to the postpartum ward. While this may be more efficient in identifying clinically significant distress or severe pain experiences, this introduces the limitations of missing the opportunity to intervene or treat pain and recall bias, since memories could be influenced by intraoperative anxiolytic administration, neonatal outcome or maternal postpartum mood. The optimal timing and phrasing of pain assessments therefore need to be evaluated to help establish best practices.

Routine follow-up questions such as "Did you experience pain during your cesarean delivery?" have now been incorporated into the anesthesia follow-up rounds at Stanford University and in ongoing prospective research at Columbia University. This approach has proved successful at identifying women who had negative experiences during surgery, and creates a useful opportunity for debriefing and offering early psychological support during inpatient hospitalization, facilitating triage, referring to other specialties and coordinating follow-up care.

Challenges in studying treatment

Designing RCTs to evaluate and determine optimal interventions and management for intraoperative pain during cesarean delivery is inherently difficult. The difficulty stems from the broad clinical spectrum of pain during surgery. Differentiating between a transient tugging or stretching sensation that perhaps is reported as mild pain (e.g. a pain numerical rating score of 1/10), or even if moderate (5/10), but resolves within a second, may only require reassurance from the anesthesia provider. This differs significantly from persistent sharp, visceral pain reported as severe (10/10), caused by an unblocked dermatome. The psychological consequences of these two experiences are also likely to be distinctly different and therefore not directly comparable when comparing treatments that patients are randomized to receive. Furthermore, in our experience, and also reported in other studies,[14,15] some patients do not wish to receive medication even when they experience mild to moderate pain. Therefore, a RCT comparing intraoperative pharmacological treatments may risk conflating these heterogeneous scenarios. Additionally risks of general anesthesia may be justified for severe pain not responding to treatment, but risks versus benefits may not be as justified for milder pain experiences. Furthermore, ethical and logistical barriers to randomizing women in acute distress must be considered and navigated.

Questions also remain regarding how best to treat intraoperative pain and what modalities to compare. For example, general anesthesia versus sedation, or which sedative agent to recommend for anxiolysis (midazolam, dexmedetomidine, ketamine, or propofol) or which analgesic agent (fentanyl, hydromorphone, or morphine), or combinations of agents to administer in each treatment arm of a planned study. Furthermore, the timing of drug administration in a RCT (e.g. following skin incision, pre-delivery, immediately following delivery, during fascia or skin closure) may increase heterogeneity further among these complex scenarios. Recent evidence even suggests that esketamine may mitigate subsequent postnatal depression after cesarean delivery, but its potential effects on acute and post-traumatic stress disorder (PTSD) after intraoperative pain are unknown.[23] These challenges should temper our expectations for definitive RCTs and instead encourage mixed-method studies and pragmatic trial approaches.

When assessing the long-term psychological impact of intraoperative pain, many confounding factors (e.g. preoperative adverse childhood experiences, anxiety or depression) need to be controlled for as these may also impact the prevalence and long-term sequalae from intraoperative pain. Additionally, the potential negative psychological impacts of treatment (e.g. general anesthesia resulting in a woman missing the birth of her child) need to be compared with other treatment modalities.

Future directions

Addressing intraoperative pain during cesarean delivery requires a multipronged agenda that spans counseling, intraoperative management, follow-up, and systems redesign. Several of these areas have been covered by the OAA and the French bulletin in the form of expert opinion, but recommendations based on high quality evidence are still lacking.[6,7]

Below are several clinical recommendations that should be considered by anesthesia care providers based on contemporary best available evidence:

1. **Focus on prevention:** the highest prevalence of intraoperative pain is with labor epidural conversion for cesarean delivery anesthesia. Protocols and strategies that focus on optimizing and/or replacing labor epidurals that are poorly functioning is essential. Block testing that demonstrates adequate anesthesia prior to incision is critical, and appropriate clinician education to ensure best practices is crucial.
2. **Preoperative counseling:** patients should be informed that they will feel sensations, some of which may be uncomfortable, including “pulling and intense movement”, and that current research reports approximately 15 % experience pain or discomfort despite optimal neuraxial anesthesia, and that this can be promptly treated. Resources such as the SOAP patient-counseling toolkit can aid consistent messaging and verbiage.[4]
3. **Paradigm shift:** anesthesiologists need to understand that telling women, “pressure is normal, but you will not feel pain” is not accurately preparing women for intraoperative sensations, and when repeated intraoperatively, this may invalidate patient experiences. For some women, discriminating between pressure and pain is difficult, and pressure may be painful or unbearable. Therefore, a better question may be to ask women if they would like medication to improve their comfort. (“Are you uncomfortable? Would you like us to give you some medication to help you with these sensations,” irrespective of whether it is pressure, pulling/tugging or pain, and regardless of the pain score).
4. **Identification of clinical phenotypes associated with worse pain:** studies are needed to determine which patterns of intraoperative pain negatively impact postpartum recovery most, and those associated with postpartum psychiatric disorders. This has relevance, given the worsening trends in US maternal mortality.
5. **Bundled care interventions:** combining optimized neuraxial techniques, intravenous pharmacological interventions (e.g. midazolam, dexmedetomidine and fentanyl), environmental modification (e.g. patient-selected music), and immediate psychological debriefing may improve both physical and emotional outcomes. The optimum pharmacological stepwise or bundled approach would need careful consideration and consensus prior to implementation, and may depend on individual institutional resources and practices.
6. **Comprehensive documentation:** clear recording of block characteristics, timing of pain episodes, and interventions would facilitate benchmarking and research, and addresses the problem of inadequate documentation that frequently accompanies cases of intraoperative pain during cesarean delivery.[24]
7. **Structured follow-up**: quality-improvement programs could incorporate standardized pain and satisfaction assessments intraoperatively, in the PACU, and on postpartum day 1–2, depending on staffing availability. Patients experiencing pain should ideally debrief with an anesthesiologist and be referred to psychological services for further follow up if deemed necessary.

Suggested key research priority areas regarding intraoperative pain are summarized in Table 1.

Reflections and next steps

The triangulation of findings from both quantitative and qualitative investigations marks an inflection point in improved knowledge and understanding surrounding intraoperative pain during cesarean delivery. We now appreciate that the “failure” of neuraxial anesthesia cannot be solely measured by the need for supplemental drugs or conversion to general anesthesia, whether based on a pain score or not. Rather, the measure must encompass the patient’s subjective experience, the “phenomenon”, which is influenced by many complex domains, many without measurable outcomes. Many patient-reported domains are potentially modifiable in nature, and need to be the focus of future research and clinical care practices. The specialty of obstetric anesthesiology must continue to lead this conversation, and develop strategies for integrating patient-reported experiences into metrics of quality and safety, which can be used to benchmark practice.

Conclusion

Intraoperative pain during cesarean delivery represents an unresolved problem from both a clinical and research standpoint. This topic is of great public health significance since cesarean delivery is the most common inpatient surgery performed globally. However, solving intraoperative pain may prove very difficult, and the complexity of this construct defies simple solutions. Quantitative data have defined its prevalence and risk factors; qualitative insights have revealed its meaning. The next step is to bridge these findings through patient-

Table 1
Suggested research priorities regarding intraoperative pain during cesarean delivery.

Topic	Explanation or comments
Prediction models	Preoperative screening for anxiety, pain catastrophizing, and adverse childhood experiences among other maternal variables, could help identify women at higher risk of intraoperative pain. This could help guide individualized counseling or consideration of preoperative anxiolysis from the outset to minimize the risk of traumatic childbirth.
Risk-stratified choice of anesthesia	Future studies should evaluate whether certain patients may benefit from elective general anesthesia when the predicted risk of intraoperative pain or psychological harm is high. Reliable, accurate and generalizable risk prediction models would however need to be developed and validated prior to instituting such studies.
Intraoperative communication	Studies should explore the optimal frequency and style of inquiry regarding pain, including verbal versus non-verbal cues, patient-initiated versus clinician-initiated; and measure both satisfaction with pain management, anesthesia care, and associated PTSD and anxiety outcomes.
Referral pathways	Development of standardized criteria for mental-health referral following reported intraoperative pain is urgently needed. PCL-5 is not specific for acute stress, and PTSD can only be diagnosed after symptoms are experienced for more than 1 month. Optimal postpartum screening and referral practices need to be developed to minimize heterogeneity in clinical practice.
Psychological sequelae	The association between intraoperative pain and postnatal depression, anxiety, and PTSD requires longitudinal, multicenter study using optimal and validated instruments such as the Edinburgh Postnatal Depression Scale and PCL-5.
Collaborative research infrastructure	Given the incidence of severe pain and the breadth of its potential sequelae, multicenter collaboration is essential. Dedicated funding streams recognizing maternal psychological outcomes as legitimate endpoints must follow.

PTSD = Post Traumatic Stress Disorder; PCL-5 = The PTSD Checklist for the DSM-5.

centered care, research, education, quality assurance and policy. As has been written before regarding intraoperative pain management, *divinum sedare dolorem*, it is divine to alleviate pain. To do so effectively, we must listen to and learn from those who experience it.

Authors contribution

PS wrote and edited this editorial.
BC helped write and edit this editorial.
RL helped write and edit this editorial.

Consent for publication

Authors consent for this manuscript to be published as per journal rules.

Availability of data and materials

Funding: None relevant to this editorial.

CRediT authorship contribution statement

Pervez Sultan: Writing – review & editing, Writing – original draft, Conceptualization. **Brendan Carvalho:** Writing – review & editing. **Ruth Landau:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Pervez Sultan and Brendan Carvalho were co-authors on the article by Callihan et al. discussed in this Editorial, but report no other conflicts of interest. Ruth Landau is the Editor in Chief of the *International Journal of Obstetric Anesthesia.*

This manuscript was handled by Prof. C. Weiniger (*IJOA* Editor).

Acknowledgment

Not applicable.

References

1. Charles EA, Carter H, Stanford S, et al. Intraoperative pain during cesarean delivery under neuraxial anesthesia: a systematic review and meta-analysis. *Anesthesiology*. 2025. https://doi.org/10.1097/ALN.0000000000005486.
2. National center for health statistics: Method of delivery. Centers for Disease Control and Prevention. https://www.cdc.gov/nchs/fastats/delivery.htm [accessed October 1, 2025].
3. Zakowski M, Fardelmann K, Hofkamp M. Pain during cesarean delivery: we can and must do better. *Anesthesiology*. 2024;140(6):1236–1237. https://doi.org/10.1097/ALN.0000000000004968.
4. Society for Obstetric Anesthesia and Perinatology. Peripartum Pain Management Toolkit. https://www.soap.org/infographics [accessed October 7, 2025].
5. The New York Times. The retrievals podcast. Season 2. https://www.nytimes.com/2025/07/10/podcasts/the-retrievals-c-section-women-health.html?action=-click&module=audio-series-bar&region=header&pgtype=Article [accessed October 10, 2025].
6. Plaat F, Stanford S, Lucas D, et al. Prevention and management of intra-operative pain during caesarean section under neuraxial anaesthesia: a technical and interpersonal approach. *Anaesthesia*. 2022;77(5):588–597. https://doi.org/10.1111/anae.15717.
7. Keita H, Deruelle P, Bouvet L, et al. Raising awareness to prevent, recognise and manage acute pain during caesarean delivery: the French Practice Bulletin. *Anaesth Crit Care Pain Med*. 2021;40(5), 100934. https://doi.org/10.1016/j.accpm.2021.100934.
8. Patel R, Kua J, Sharawi N, et al. Inadequate neuraxial anaesthesia in patients undergoing elective caesarean section: a systematic review. *Anaesthesia*. 2022;77(5):598–604. https://doi.org/10.1111/anae.15657.
9. Stanford S. What is 'genuine' failure of neuraxial anaesthesia? *Anaesthesia*. 2022;77:523–526. https://doi.org/10.1111/anae.15723.
10. Sultan P, Patel R, Sharawi N, Moonesinghe S. Divinum sedare dolorem: it is divine to alleviate pain. *Anaesthesia*. 2022;77(8):942–943. https://doi.org/10.1111/anae.15743.
11. Callihan P, Cottrell Bekemeyer ZJ, Pedreira R, et al. Intraoperative pain during cesarean delivery: a qualitative concept elicitation study. *Int J Obstet Anesth*. 2025;64, 104748. https://doi.org/10.1016/j.ijoa.2025.104748.
12. Kinsella S. A prospective audit of regional anaesthesia failure in 5080 Caesarean sections. *Anaesthesia*. 2008;63(8):822–832. https://doi.org/10.1111/j.1365-2044.2008.05499.x.
13. Pan PH, Bogard TD, Owen MD. Incidence and characteristics of failures in obstetric neuraxial analgesia and anesthesia: a retrospective analysis of 19,259 deliveries. *Int J Obstet Anesth*. 2004;13(4):227–233. https://doi.org/10.1016/j.ijoa.2004.04.008.
14. Keltz A, Heesen P, Katz D, et al. Intraoperative pain during caesarean delivery: Incidence, risk factors and physician perception. *Eur J Pain*. 2022;26(1):219–226. https://doi.org/10.1002/ejp.1856.
15. Sanchez J, Prabhu R, Guglielminotti J, Landau R. Pain during cesarean delivery: a patient-related prospective observational study assessing the incidence and risk factors for intraoperative pain and intravenous medication administration. *Anaesth Crit Care Pain Med*. 2024;43(1), 101310. https://doi.org/10.1016/j.accpm.2023.101310.
16. Orbach-Zinger S, Olliges E, Garren A, et al. Patient/anesthesiologist intersubjective experiences and intravenous supplementation during elective cesarean delivery: a prospective patient-reported outcome study. *J Clin Anesth*. 2025;100, 111689. https://doi.org/10.1016/j.jclinane.2024.111689.
17. Drake-Brockman TFE, Dodd M, Cooney S, et al. A qualitative exploration of the phenomenology of pain in children to inform pain assessment methods. *PLOS One*. 2025;20(9), e0332570. https://doi.org/10.1371/journal.pone.0332570.
18. Esparza Á, Garcia L, López Muñoz F, Olano A, García-Rico EM. Suffering and the collapse of language: between the sayable and the impossible. *Med Humanit* 2025: medhum-2025-013361. Doi: 10.1136/medhum-2025-013361.
19. Jiang A, Perry T, Walker K, Burfoot A, Patterson L. Surgical sensation during caesarean section: a qualitative analysis. *Int J Obstet Anesth*. 2024;57, 103935. https://doi.org/10.1016/j.ijoa.2023.103935.
20. Brogly N, Valbuena Gómez I, Afshari A, et al. ESAIC focused guidelines for the management of the failing epidural during labour epidural analgesia. *Eur J Anaesthesiol*. 2025;42(2):96–112. https://doi.org/10.1097/EJA.0000000000002108.

21. Varelmann D, Pancaro C, Cappiello EC, Camann WR. Nocebo-induced hyperalgesia during local anesthetic injection. *Anesth Analg.* 2010;110(3):868–870. https://doi.org/10.1213/ANE.0b013e3181cc5727.
22. Van Dorp TIB, Kwant M, Lameijer H. Could we use the comfort score instead of the numeric rating pain score? a proof of concept pilot study. *Am J Emerg Med.* 2023;68: 64–67. https://doi.org/10.1016/j.ajem.2023.01.015.
23. Li Y, Zhao K, Cao P, et al. Perioperative use of esketamine for the prevention of postpartum depression after cesarean section: a meta-analysis 2024. Doi: 10.2139/ssrn.4861065.
24. McCombe K, Bogod DG. Learning from the law: a review of 21 years of litigation for anaesthetic negligence resulting in peripartum hypoxic ischaemic encephalopathy. *Anaesthesia.* 2022;77(8):919–928. https://doi.org/10.1111/anae.15741.

Pervez Sultan[a,*], Brendan Carvalho[b], Ruth Landau[c]

[a] *Department of Anesthesiology, Perioperative and Pain Medicine, Stanford University School of Medicine, CA, USA*

[b] *Department of Anesthesiology, Perioperative and Pain medicine, Stanford University School of Medicine, CA, USA*

[c] *Department of Anesthesiology, Columbia University, NY, USA*

[*] Corresponding author at: Department of Anesthesiology, Perioperative and Pain Medicine, Stanford University School of Medicine, 300 Pasteur Drive, Stanford, CA 94305, USA.

E-mail address: psultan@stanford.edu (P. Sultan).